

Statistical mechanical theory of binding-induced stresses and consequences for biological activity of nanoparticles

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Ligand-functionalised nanoparticles and polymers are widely studied for applications ranging from drug delivery to antiviral systems. In these contexts, a central but often overlooked quantity is the mechanical force that ligands collectively exert on the receptors they bind, since forces above certain thresholds can trigger downstream biological processes including receptor extraction, mechanotransduction, and structural disruption of target surfaces. Here we present a theoretical model that links the physical properties of multivalent vectors, including ligand grafting density, tether length, and nanoparticle size, to the average force they induce on a target surface. Using this model, we identify strategies to maximise this force and show that, for typical biological ligand–receptor pairs and nanoparticle dimensions, its predicted magnitude falls in the piconewton range characteristic of key biological force-driven processes, from molecular motor activity and immune cell antigen extraction to integrin-mediated mechanosensing. Under favourable conditions, the force can be made large enough to significantly accelerate the extraction of transmembrane proteins from a lipid bilayer. These results provide a quantitative framework for understanding the mechanical action of nanoparticles on biological surfaces such as viral envelopes and cell membranes, and offer a new perspective on both cytotoxicity mechanisms and the rational design of mechanically active antiviral nanoparticles.

Nanoparticles and polymers functionalised with multiple ligands, commonly referred to as multivalent systems, have been studied for decades for their ability to interact with, and potentially modulate, biological systems. Inspiration comes directly from nature: cells, viruses, and bacteria all communicate through the formation of multiple ligand–receptor bonds, often in a concerted fashion [1, 2].

A prominent application of multivalent nanoparticles is as antiviral agents [3, 4]. Early work on the subject focused on systems that bind to viral surface receptors via ligand–receptor interactions, thereby preventing the virus from engaging its cognate cellular receptors. The nanoparticle ligands occupy viral binding sites while the shell of bound nanoparticles sterically shields the virus from the cell surface, jointly inhibiting viral adsorption and subsequent endocytic uptake. A theoretical model by Liese *et al.* [5] based on this mechanism has been shown to capture experimental results well, and antivirals operating in this way are termed virustatic. In 2018, Cagno *et al.* [6] reported a qualitatively distinct class of system, which they termed virucidal nanoparticles. Unlike virustatic agents, virucidal nanoparticles not only prevent viral adsorption but induce disassembly, and thus irreversible inactivation of the virus. To the best of our knowledge, no microscopic mechanism has been clearly validated to explain this virucidal activity.

In addition to viruses, multivalent systems can inter-

act with receptors on eukaryotic cell surfaces for different purposes. Upon binding, nanoparticles may be endocytosed or become partially engulfed at the membrane. In either case, ligand–receptor engagement can drive receptor clustering and, potentially, receptor extraction from the membrane. Such processes have been linked to immune activation: for example, ligand–receptor forces are central to the mechanical proofreading mechanism by which B cells select high-affinity antigens during affinity maturation. More generally, mechanical forces transmitted through ligand–receptor bonds are exploited by cells to activate downstream signalling once a certain threshold is exceeded [7].

Crucially, the magnitude of these forces is not solely determined by the specific ligand–receptor pair, but also by the multivalent presentation geometry, *i.e.*, the relative spatial arrangement of ligands and receptors. This geometry determines the magnitude of combinatorial entropic energy contributions by modulating the number of accessible ligand–receptor bonding patterns.

Bonds requiring excessive ligand stretching generate strong stresses but are rare and short-lived, while bonds that form with negligible stretching have higher occurrences but correspondingly exert smaller forces. Which bonds form at any given time is governed by a dynamic equilibrium that depends on the relative positions of ligands and receptors. Understanding how these forces depend on experimentally tuneable nanoparticle properties is therefore essential for rationalising their downstream biological effects.

For this reason, in this letter we derive a theoretical model, based on the statistical mechanical theory of

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multivalent interactions [8–11], to calculate the average force exerted on a surface-grafted receptor by a ligand-coated multivalent nanoparticle. We show that in accessible regions of the nanoparticle design space, the estimated force magnitude is consistent with the mechanical stresses that can induce mechanosensing and might disrupt viral capsid assembly. We also identify design principles for maximising or minimising this force depending on the desired outcome: enhancing virucidal activity, activating mechanosensitive signalling pathways to promote endocytosis, or minimising immunogenicity by suppressing mechanical perturbations.

From a general point of view, the scenario we aim to describe is the following. A surface coated with receptors is immersed in a solution containing nanoparticles coated with N_L^{tot} ligands per nanoparticle. This solution could be, for example, the blood plasma, where nanoparticles are injected to fight a viral infection. The surface receptors can be either disposed randomly on the surface, or follow a certain regular pattern, characterised by a 2D pair correlation function $g(\mathbf{r})$ [12]. For ease of calculation, $g(\mathbf{r})$ will be assumed isotropic, in mathematical notation $g(\mathbf{r}) \rightarrow g(r)$, but similar formulas apply also for the case where directionality matters (for example, in the study of viruses such as adenovirus or poliovirus where the viral capsid can be thought of as a regular, icosahedric distribution of receptors). At each point in time, there will be a subset $N_L < N_L^{\text{tot}}$ ligands facing the surface and being able to potentially form bonds with a subset $N_R < N_R^{\text{tot}}$ of the receptors. The bond strength is characterised by the number $\chi = \exp(-\beta\Delta G_{\text{bond}})$, where ΔG_{bond} is the bond energy and $\beta = 1/k_B T$ is the inverse thermal energy, where k_B is Boltzmann's constant and T the absolute temperature of the system. Note that, as thoroughly described by Varilly *et al* [8], the bond energy is a function of the relative position of ligands and receptors, as well as dependent on the specific properties of the linker used to graft the ligands on the nanoparticles and can in general be different for each pair. Upon formation of a bond with a ligand, depending on the exact geometry of the ligand-receptor bond, a receptor $i \in N_R$ on the surface will experience a certain force $\mathbf{F}_i$. Because for the kind of ligand-receptor pairs typically used in these systems, bonds are non-covalent in nature and thus reversible, bonds will form and break, and as a result, this force will fluctuate in time. We seek to find an expression to calculate the average of this force under all potential patterns of ligand-receptor bonds that can be formed.

The starting point for this calculation is a formula for the force that two binders i and j exert on each other at thermodynamic equilibrium. A binder is either a ligand or a receptor, but the distinction between them is arbitrary. Purely for clarity, here we use the term ligand for a binder on a nanoparticle and receptor for a binder on a surface, but the theory is agnostic to this definition and, as long as i and j can form bonds between each other, their distinction between ligand and receptor is irrelevant. Having said that, it was previously shown that

the force between i and j is given by the solution of the following set of coupled equations:

$$p_i + \sum_j p_{ij} = 1 \quad (1)$$

$$p_{ij} = p_i p_j \chi_{ij} \quad (2)$$

$$\mathbf{F}_i = -p_{ij} \nabla_i \Delta G(\mathbf{r}_{ij}) \quad (3)$$

This equation is exact in the case of mobile receptors and a large number of binders, but is also retrieved as a mean-field approximation for fixed receptors (and a large number of binders) upon substituting the fraction of binders i involved in n_{ij} bonds with the probability to form a bond between i and j , as can be evinced by comparing the identical form of the partition function in Ref. [13] and [8],[9]. Note that Eqs. 1, 2 constitute a set of coupled equations since there is one for every distinguishable binder in the system, and they have to be solved self-consistently [9]. A Python implementation computing the solution of these equations can be found at <https://github.com/patvarilly/DNACC>.

We now turn to specialise the previous equations to our scenario. For the sake of simplicity, we consider the case of nanoparticles with relatively long ligands compared to their core so that the exact grafting position on the nanoparticle surface does not matter and all ligands are interchangeable in our description, that is, $\mathbf{r}_{i,L} = \mathbf{r}_{\text{NP}} + \delta\mathbf{r}_i \approx \mathbf{r}_{\text{NP}}$, where $\mathbf{r}_i$ is the relative position of the grafting point on the surface with respect to the nanoparticle centre, see Figure 1 for a schematic illustration of the system. This approximation becomes more and more correct as $R_{\text{NP}}/R_{\text{ee}} \rightarrow 0$, where R_{ee} , the end-to-end distance of the polymer tethering the ligand to the nanoparticle's surface (often used as a measure of the polymer size), and R_{NP} the radius of the nanoparticle.

Specialising Eq. 1, Eq. 2 to our system we obtain:

$$p_L(z) + \int_0^\infty 2\pi r \sigma_R(r) \frac{\chi(r, z) p_L(z)}{1 + N_L p_L(z) \chi(r, z)} dr = 1 \quad (4)$$

$$p_R(r, z) = \frac{1}{1 + N_L p_L(z) \chi(r, z)} \quad (5)$$

For a single receptor

$$\beta F_{i,\alpha} = N_L p_i(r, z) p_L(z) \nabla_\alpha \chi(r, z) \quad (6)$$

where $\alpha \in \{r, z\}$ denotes the radial or vertical component, and ∇_α denotes the corresponding component of the gradient operator.

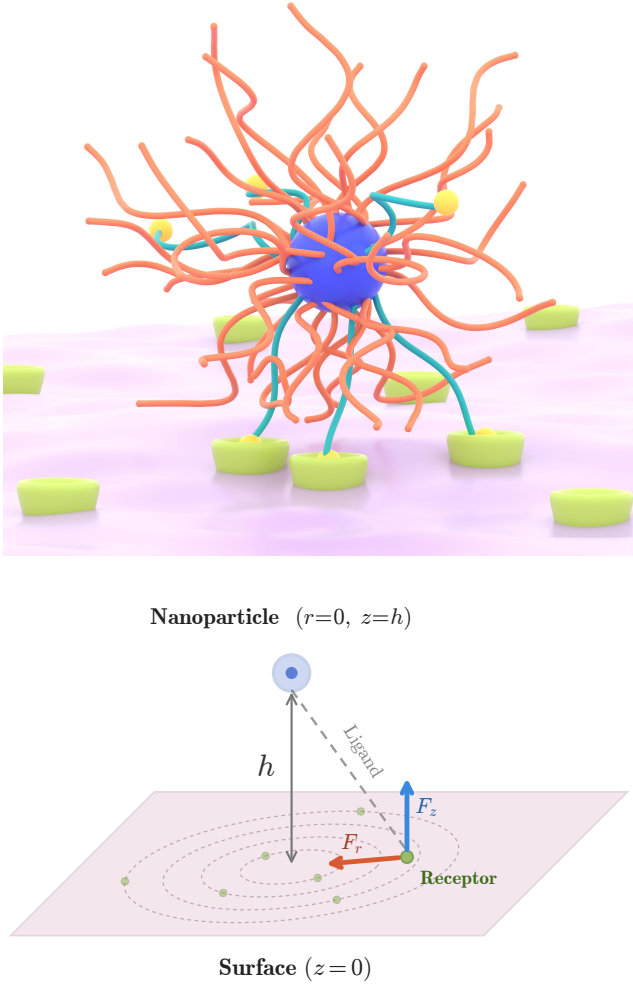


Figure 1: **Schematic representation of our system.** A nanoparticle (blue sphere) coated with functional ligands (cyan tethers and yellow spheres) and inert polymers (orange chains) interacts with a receptor-coated surface (green discs). While ligand-receptor binding provides the attractive component, excluded-volume interactions (for instance, tether or ligand confinement due to proximity with the target surface) contribute an extra repulsive term. The accompanying diagram illustrates the geometric setup of the theoretical model: the nanoparticle centre is located at height $z = h$ above the surface, and receptors sit at radial positions from the point directly beneath it. Each ligand-receptor bond exerts a force that is decomposed into a radial component F_r (parallel to the surface) and a vertical component F_z (perpendicular to the surface).

Averaging over all receptors in the system, we obtain:

$$F_\alpha = \frac{f}{N_R} \int_0^{r_{\max}} 2\pi r \sigma_R(r) F_b(r) dr \quad (7)$$

where $N_R = \int_0^{r_{\max}} 2\pi r \sigma_R(r) dr$ is the number of receptors the particle can bind to and $r_{\max}$ the maximum extension of the ligand, and $F_b(r)$ the average force on receptors at position r , where the Boltzmann-weighted average is over

all nanoparticle potential heights z , given by:

$$\beta F_b(r) = \int_0^{z_{\max}} N_L p_L(z) p_R(r, z) \nabla_\alpha \chi(r, z) p(z|B) dz \quad (8)$$

where $z_{\max}$ is the maximum nanoparticle height at which bonds can still be formed with the receptors, and $p(z|B)$ is the conditional probability of finding the particle at position z , when bonds are allowed to form (which we will refer to as the bound state [9], B), given by:

$$p(z|B) = \frac{\exp[-\beta A(z)]}{\Omega} \quad (9)$$

where

$$\Omega = \int_0^{z_{\max}} \exp[-\beta A(z)] dz \quad (10)$$

is the partition function of the nanoparticle in the bound state $p(z|B)$, and $A(z)$ is the free-energy of interaction between a particle and its binding site as a function of the height z , see Figure 1 for reference. We will discuss the exact form of this term later. Going back to Eq. 7, f is the fraction of potential adsorption sites in the system with a particle bound to them, given by:

$$f = \frac{[(c_{NP} + c_S) \tilde{K} + 1]}{2 \tilde{K} c_S} \times \left(1 - \sqrt{1 - \frac{4 \tilde{K}^2 c_{NP} c_S}{[(c_{NP} + c_S) \tilde{K} + 1]^2}} \right), \quad (11)$$

where $c_S \propto c_T \cdot (R_T/R_{NP})^2$ is the total number of adsorption sites in the system, with c_T , R_T and R_{NP} being the molar concentration of targets (viruses or cells), the radius of the target and the radius of a nanoparticle in the system (all assumed to be spherical). The value of the effective nanoparticle/binding site binding constant is given by:

$$\tilde{K} = v_{\text{bind}} \exp(-\beta \epsilon) \quad (12)$$

where $v_{\text{bind}} = A_s z_{\max}$ is the binding volume of the nanoparticle, that is, the region within which the nanoparticle is considered bound to the virus, A_s the effective area of a single adsorption site. ϵ is the effective value of the binding energy, defined as:

$$\beta \epsilon = -\log \frac{\Omega}{z_{\max}} \quad (13)$$

or written in a more compact form one simply has:

$$\tilde{K} = A_s \Omega \quad (14)$$

The fraction of bound sites is simply calculated by describing binding between a nanoparticle and an adsorption site as a chemical reaction at equilibrium $\text{NP} + \text{S} \rightleftharpoons \text{NPS}$.

In practice, Eq. 7 provides the average force on a receptor, calculated by taking an average over all receptors in the system and over all potential positions of a bound nanoparticle. In particular, this formula takes into account the fact that not all receptors might have a nanoparticle within reach, and thus some of the receptors in the system might not feel any force. For example, when the number of nanoparticles in the system is much smaller than the total number of potential binding sites present, no matter how strong the force on a bound receptor is, the average force might still be small because only a few receptors will feel it. We note that, when the goal is to predict the force exerted by a single nanoparticle on a single, specific receptor, the correct measure to use is F_b alone, the average force on a receptor residing on an adsorption site with a bound nanoparticle on it. However, when the goal is to predict the effect of a solution of nanoparticles on the entire system, the average over all receptors implied in Eq. 7 is more appropriate.

Let us now go back to the exact definition of $A(z)$, the interaction energy between the nanoparticle and the binding site. Without loss of generality, we can split this energy into three contributions:

$$A(z) = A_{\text{bond}}(z) + A_{\text{steric}}(z) + A_{\text{ns}}(z) \quad (15)$$

The first contribution is the attractive interaction A_{bond} due to the formation of ligand-receptor bonds. For this system, the bond energy is given under the same approximations used to calculate the force by:

$$\begin{aligned} \beta A_{\text{bond}}(z) = & N_L \left[\ln(p_L(z)) + \frac{1}{2}(1 - p_L(z)) \right] \\ & + \int_0^\infty 2\pi r \sigma_R(r) \left[\ln(p_R(r, z)) + \frac{1}{2}(1 - p_R(r, z)) \right] dr \end{aligned} \quad (16)$$

For a system of ligand-receptor bonds, there is also generally a second term steric interaction between the ligands/receptors and the surfaces of the nanoparticle and of the virus, which cannot be penetrated. As discussed in [8], this excluded volume contribution is proportional to the number of ligands and is generally given by:

$$\beta A_{\text{steric}}(z) = -N_L \ln \left(\frac{Q(z)}{Q(\infty)} \right), \quad (17)$$

where $Q(z)$ is the molecular partition function of the ligand at a distance z from the surface [8]. A contribution

of exactly the same form can also be modelled for receptors on the virus surface but here we only consider the ligand contribution for compactness as the same general results would be observed for the receptor-dependent repulsion. To further validate the robustness of our results, in the supplementary information the effect of assuming a different functional form for the repulsive energy term A_{steric} is also investigated, showing the same qualitative trends described in the main text are retrieved.

Finally, we can dump all other non-specific interactions into a third term A_{ns} . This third term can be attractive or repulsive, depending on its exact physical origins. In this work we will assume this term to be zero (which should be true for most well-designed systems of nanoparticles where such non-specific interactions must always be minimised), but our theoretical framework can be readily extended to include the effect of various types of non-specific interactions.

Whereas up to this point the theory is relatively general, making use of the previous expression requires specifying a coarse-grained model for the ligands, the receptors, and their interaction. Here, we will use a relatively general and widely used model presented by Varilly *et al* in Ref. [8]. The interested reader is strongly suggested to find all details there, here they will be summarised very briefly. In this model, the single bond energy between a ligand and a receptor can be expressed as the sum of two terms:

$$\Delta G_{ij} = \Delta G_{ij}^0 + \Delta G^{\text{cnf}}(\mathbf{r}_i, \mathbf{r}_j) \quad (18)$$

i.e., the term ΔG_{ij}^0 dependent on the free-energy of binding for *free* ligands and receptor in solution, related to the experimentally measurable equilibrium binding constant for that specific pair (i, j) , and a term $\Delta G^{\text{cnf}}(\mathbf{r}_i, \mathbf{r}_j)$ depending on the properties of the molecular construct used to graft those ligand and receptor to the nanoparticle or the surface. Although the precise functional form of the latter term is specific to any given ligand-receptor pair and surface tethering of binders, for a qualitative discussion, as done in Refs. [5, 14] we can simply approximate it to a second order expansion around the equilibrium distance x_{eq} , giving the general harmonic form:

$$\Delta G^{\text{cnf}}(\mathbf{r}_i, \mathbf{r}_j) = \frac{1}{2} k_{\text{eff}} (|\mathbf{r}_i - \mathbf{r}_j| - x_{\text{eq}})^2 \quad (19)$$

where $r_{ij} = |\mathbf{r}_i - \mathbf{r}_j|$ is the distance between the ligand and receptor grafting points and k_{eff} is an effective spring constant. In order to later provide a simple order of magnitude estimate of the forces involved, here we will treat each ligand as an ideal polymer, resulting in $k_{\text{eff}} = 3k_B T / R_{ee}^2$ and $x_{\text{eq}} = 0$, where $\sqrt{N}a$ is the average end-to-end distance of the tether, N the number of Kuhn segments in the polymer, and a the Kuhn length, which for very flexible polymers is around twice

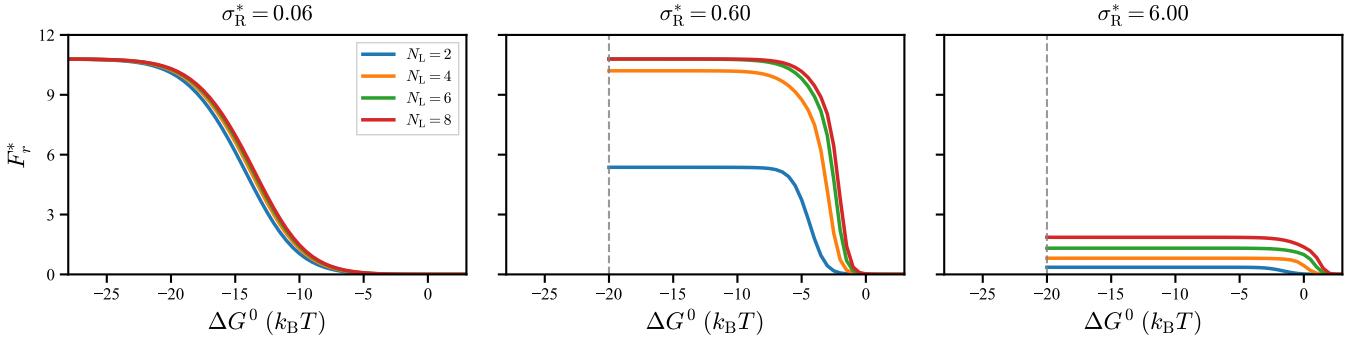


Figure 2: **Radial force on a bound receptor as a function of binding strength.** Dimensionless average radial force F_r^* exerted on a receptor as a function of the intrinsic ligand-receptor binding free energy ΔG^0 , for different numbers of ligands $N_L = [2, 4, 6, 8]$ and three dimensionless receptor densities: $\sigma_R^* = 0.06$ (left), $\sigma_R^* = 0.60$ (middle), and $\sigma_R^* = 6.00$ (right). As explained in the main text, all quantities are presented in dimensionless form: lengths are scaled by the ligand tether end-to-end distance $R_{ee} = \sqrt{N}a$, energies by the thermal energy $k_B T$ and forces by $k_B T/R_{ee}$; in particular, $F_r^* = F_r R_{ee}/k_B T$ and $\sigma_R^* = \sigma_R R_{ee}^2$. At low receptor density, the curves for different N_L nearly collapse onto a single one, whereas at higher densities the plateau force grows with N_L and the onset of binding shifts towards weaker affinities.

the monomer length. For the same case, the steric interaction is given by: $-N_L k_B T \ln [Z(z)/Z(\infty)]$, where $Z(x) = \int_{-x}^{\infty} \exp[-z'^2/(Na^2)] dz'$. Receptors are treated as rigid with a fixed structure, as one would expect for example for well-structured globular proteins due to their strong network of hydrogen bonds.

Here and throughout, for the sake of generality, all quantities are presented in dimensionless form: lengths are scaled by the ligand end to end distance R_{ee} , energies by $k_B T$ and forces by $k_B T/R_{ee}$; in particular, $F_r^* = F_r R_{ee}/k_B T$ and $\sigma_R^* = \sigma_R R_{ee}^2$. Unless stated otherwise, as a reference we will also use $x_{eq} = 0$, $N_L = 5$ and $N_{steric} = 0$, where the latter is the number of inactive ligands in the system, that is, ligands that can only provide steric protection but cannot bind to the receptor. These numbers can be easily changed to readily describe any arbitrary system. We will also see later what these parameters' typical values correspond to when typical numbers for a standard nanoparticle system are used. Finally, both for simplicity and because of its biological relevance for most receptors at their typical expression level, a random disposition of receptors is assumed, *i.e.*, we set a uniform value for their surface density $\sigma(r) = \sigma$. Deviations from a uniform density, which we do not expect to change qualitatively the general trends presented here, will be discussed later.

With these choices, in Figure 2, the effect of multivalency is investigated by studying how the average force in the radial direction increases as a function of the bond energy for different values of the number of ligands on the nanoparticle and different densities of receptors. Qualitatively identical trends are obtained for the average force in the direction perpendicular to the receptors plane, as can be seen in the Supplementary Information.

Notably, the force has a strongly non-linear behaviour with the bond energy, and rapidly increases from zero to a plateau value after the bond strength, *i.e.*, $K_{bind} =$

$e^{-\Delta G^*}$, exceeds a certain threshold.

Moreover, we observe that both the absolute value of this threshold energy and the plateau force increase with decreasing receptor density.

The reason behind the observed behaviour is readily explained by analysing the length-scales involved. The average inter-receptor distance is about $d_{R-R}^* \approx 1/\sqrt{\sigma_R^*}$. Decreasing σ_R^* therefore increases the typical ligand extension required to reach a binding partner. Within the ideal-chain picture, this raises the conformational cost ΔG^{cnf} associated with bond formation, so that binding becomes favourable only for a more negative intrinsic binding energy ΔG^0 , thereby shifting the onset to more negative values. At the same time, when $\Delta G^* \rightarrow -\infty$ and all ligands are in a bound state, the larger typical extensions imply a larger entropic elastic force per bond, consistent with the higher plateau force observed at lower σ_R^* in Figure 2.

Increasing the ligand number N_L also shifts the threshold bond energy to weaker bonds. This effect arises because a larger number of ligands increases the mean number of bonds that can form simultaneously. Multiple weaker bonds can collectively stabilise binding of the nanoparticle (as measured by the factor f in Eq. r7) at lower bond strength, hence the lower threshold energy for a finite force to appear.

A second, more complex behaviour, strongly dependent on the *scaled* receptor density σ^* , is observed, regarding the effect of the ligands number on the value of the force at plateau. For large densities, the plateau seems to increase almost linearly with N_L . At intermediate values, a somewhat mixed behaviour appears, with the force increasing first linearly with N_L , and then plateauing. At even lower densities, the same plateau value is observed, regardless of N_L . To better

Fig2b.pdf

Figure 3: **Maximum (plateau) force as a function of the number of ligands, for different densities.** Different regimes are observed, with a linear behaviour below a threshold value, and sublinear above it. Vertical lines are the threshold value calculated using our theoretical argument, corresponding to $N_L^* = \sigma_R^* \pi$, see also the main text.

analyse this behaviour, we plot in Fig. 3 the maximum value of the force, corresponding to the plateau value measured for $\Delta G^* \rightarrow -\infty$, as a function of N_L , using a double logarithmic. Different curves represent different values of the density σ^* . Reported in this way, one can better observe that, for each density, there is a threshold value for the number of ligands, let us call it N_L^* be-

low which the behaviour of the force is almost perfectly linear (slope ≈ 1), and above which a sublinear behaviour is observed. To explain why, we use the following argument: whenever possible, given their lower entropic cost, ligands will form bonds with the closest receptors. In the region of the receptor-bearing surface closest to the nanoparticle, and thus right below it, there

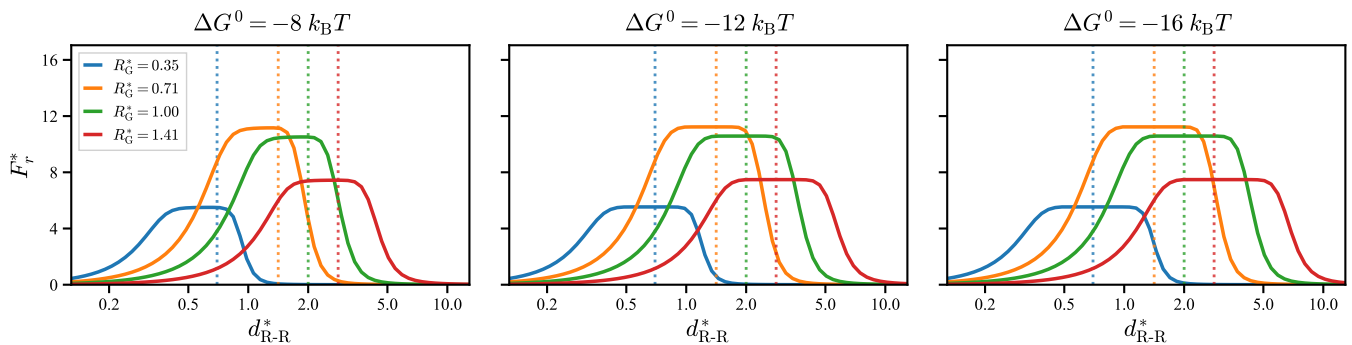


Figure 4: **This is all wrong and makes no sense, the scaling cannot be like this** Radial force on a bound receptor as a function of receptor-receptor distance. Dimensionless average radial force F_r^* as a function of the dimensionless inter-receptor distance $d_{R-R}^* = (\sigma_R^*)^{-1/2}$, for different ligand flexibilities, as characterised by the dimensionless gyration radius $R_G^* = [0.35, 0.71, 1.00, 1.41]$, and three intrinsic binding free energies: $\Delta G^0 = -8 k_B T$ (left), $-12 k_B T$ (middle), and $-16 k_B T$ (right). Vertical dotted lines mark $d_{R-R}^* = 2 R_G^*$ for each curve. The force is non-monotonic: it vanishes in both the close-packed and dilute limits, peaking at an intermediate spacing of the same scale as R_G^* . Stronger binding (left to right) broadens the plateau but leaves the peak position largely unchanged.

are approximately $N_{\text{easy}} \approx \pi R_{ee}^2 \sigma_R$ receptors available where binding requires stretching within the thermal energy ($x/R_{ee} \leq 1$). N_L^* is given by the density at which $N_L = N_{\text{easy}}$, that is, $N_L^* = \pi \sigma_R^*$ when reduced units are used. This is because all receptors can find a binding partner within this region. The vertical lines in Fig. 3 represent the value of this estimate, which capture the position of the crossover region extremely well.

With the help of Figure 4, we now turn to analyse the effect of ligand flexibility in the targeting of specific receptor densities. **Require putting reference but also carefully analysing difference with what we see here - the case previously studied was a divalent ligand, so no effect of multiple bonds beyond 2 which is the minimum needed to have any force** In this regard, we notice that previous studies have found that the maximum binding strength **but not the force, which is something different! You can have stronger binding and constant force, or no effect at all of the single-bond strength** between a multivalent construct and a receptor-bearing target has a non-monotonic behaviour and is achieved when the inter-receptor distance is almost matched (in fact, still slightly higher) to the distance between the ligands, and sharply decreases above and below the optimal distance, with a sharpness that depends on the average ligand fluctuations. We observe here a somewhat similar behaviour for the maximum force, which also displays a non-monotonic dependence on d_{R-R} with a maximum centred approximately around $2R_G$, where R_G measures the size of the polymeric ligand. However, compared to the binding strength, the force exhibits a notably broader plateau around the maximum (for the radial force), or a very broad peak (for the force in the z-direction, see the Supplementary

Information).

This non-monotonic behaviour can be justified considering the following physical mechanism: for high density of receptors, *i.e.*, low inter-receptor distance, ligands will more favourably bind those for which they can achieve the lowest effective bond energy, or in other words the one that can be bound while stretching the least, because this will have the lowest entropic penalty ΔG^{cnf} , see Eqs. 18,19. However, this means that the exerted force will also be rather low. On the opposite side, when the receptor density is very low, that is, for large inter-receptor distances, ligands have to stretch a lot to form bonds; whereas this will provide a large force, these bonded configurations will also be much less thermodynamically favourable and thus less likely, because the formation of a bond is associated with a high entropic cost. Given these considerations, it is clear that a sweet-spot should be some intermediate value of the density corresponding to an average distance that guarantees exerting enough force through stretching while also not reducing bond probability too much. In this regard, we observe that such sweetspot occurs when the distance between receptors is of the order of the typical distance spanned by the polymeric ligand due to thermal fluctuations, which is of order R_{ee} . Additionally, we can see how R_{ee} not only dictates the optimal receptor density for maximum force and its absolute value, but also the width of the force peak, which broadens for larger R_{ee} values. This should indeed be expected as longer ligands can also stretch more easily, since the effective stretching constant is inversely proportional to R_{ee}^2 , justifying the observed behaviour.

Finally, we return to the repulsive interaction term A_{steric} introduced in Eq. 15, which was previously set to zero. To assess its potential impact, Figure 5 examines a form of repulsive interaction arising from the steric interaction of inert polymer chains grafted to the nanopar-

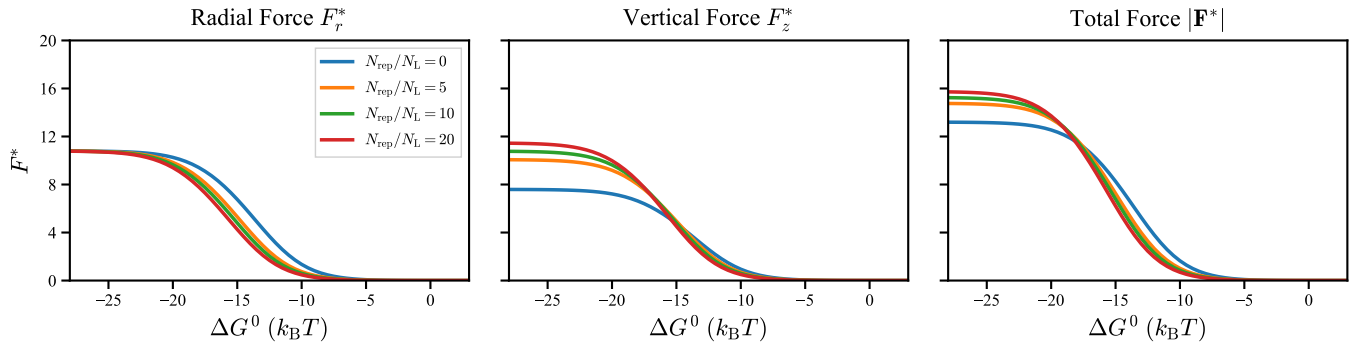


Figure 5: **Effect of inert polymer chains on the force exerted on a bound receptor.** Dimensionless radial component F_r^* (left), vertical component F_z^* (middle), and total force magnitude $|\mathbf{F}^*|$ (right) as functions of the intrinsic binding free energy ΔG^0 , for different ratios of inert (repulsive) to active (binding) polymers $N_{\text{steric}}/N_L = [0, 5, 10, 20]$ at fixed dimensionless receptor density $\sigma_R^* = 0.06$. Adding inert chains reduces the radial pulling force across the entire binding-strength range. The effect on the vertical force is non-trivial: at weak binding the inert chains slightly suppress F_z^* , whereas at strong binding they enhance the vertical pull on the receptor beyond the inert-free case, resulting in a crossover between the curves. This enhancement arises from the steric repulsion from the inert chains pushing the nanoparticle further from the surface, stretching the bound ligands and thereby increasing the extraction force on the receptor. The total force magnitude exhibits the same crossover: at weak binding the inert-free nanoparticle exerts a larger total force, while at strong binding the enhanced vertical pull more than compensates the reduced radial component, so that nanoparticles carrying more inert chains exert a larger total force on the receptor.

ticle, characterised by the ratio N_{steric}/N_L . Using polymers of the same length and rigidity as that used to graft the ligands, inert polymers give the same contribution to steric repulsion (see Eq. 17) without the additional bonding contribution. For this reason, the ratio N_{steric}/N_L has been previously exploited as a convenient way to tune the relative strength of repulsion and attraction without introducing any additional type of interaction [13, 15]. Qualitatively similar results are obtained for other forms of repulsion; see the Supplementary Information.

Adding non-specific repulsion, we observe that the radial force component F_r^* decreases with increasing N_{steric}/N_L across the entire range of ΔG^0 . The steric repulsion from the inert chains increases the average nanoparticle-surface separation, so that only receptors lying close to the point directly beneath the nanoparticle can still form bonds. The resulting reduction in the average stretching of the bound ligands, and thus of its projection on the adsorption surface, leads to a smaller average radial force on each receptor. **Need careful review** The behaviour of the vertical component F_z^* is more subtle. At weak binding, adding inert chains also suppresses F_z^* : the increased nanoparticle-surface separation raises the stretching cost of bond formation, reducing the binding probability and hence the average vertical force on the receptor, even though each individual bond that does form carries a larger average vertical load. At strong binding, the trend reverses: the binding free energy is now sufficient to maintain a high binding probability despite the increased average nanoparticle-surface separation $\langle z \rangle$. Since the mean vertical extension of

each bond grows with $\langle z \rangle$ and the stretching force increases quadratically with it, the vertical pull on each bound receptor is substantially enhanced. The enhancement of F_z^* grows with N_{steric}/N_L , producing a clear crossover between the curves. In this regard, the total force magnitude $|\mathbf{F}^*|$ inherits this crossover behaviour. At weak binding, the reduction in both components leads to a smaller total force for nanoparticles carrying more inert chains. At strong binding, the enhancement of the vertical pull outweighs the reduction in radial component, so that nanoparticles with larger N_{steric}/N_L exert a larger average total force on the receptor.

Within this framework our results suggest some general strategies to enhance the force exerted on receptors. First, the force displays a non-monotonic dependence on inter-receptor distance, peaking when $d_{R-R} \approx 2R_G$. Matching the ligand length to the receptor spacing on the target virus therefore emerges as a natural design criterion. Second, in the strong-binding regime, the inclusion of inert polymer chains can increase rather than decrease the total force by enhancing the vertical pull on each receptor, although this benefit reverses at weak binding where the inert chains reduce the total force. Importantly, all these effects can be tuned by changing the nanoparticle properties, *e.g.*, ligand number, flexibility, and density of the polymer protective shell, independently of the properties of the target receptor surface, which are typically fixed by external constraints. These design principles follow directly from the statistical mechanics of multivalent interactions and do not rely on specific molecular details of the ligand-receptor pair, sug-

gesting broad applicability across different systems and cases. Symmetrically opposite design principles apply in the case one aims at minimising the average force applied to the surface receptors in order to minimise immunogenicity and other unwanted side-effects.

We now consider the biological relevance of the force magnitudes predicted by our model. To convert our scaled quantities into physical units, we take a representative system: PLGA, a typical FDA-approved polymer used in nanoparticle formulations, as the tether. For a degree of polymerisation of 60, a monomer size of approximately 0.37 nm, and a Kuhn length of 2 nm, we obtain $R_{ee} \approx 7$ nm. At body temperature (310 K), the characteristic scale of the average force we observed $F^* = 10$ then corresponds to approximately 6 pN. This value falls squarely within the range of biologically relevant forces. Molecular motors such as kinesin and myosin generate forces of [4–6] pN, while during affinity maturation B cells exert pulling forces of 10–20 pN to extract antigens from antigen-presenting cells (APCs) [16]; T-cell receptor mechanosensing operates in the same piconewton regime. More broadly, this is the force scale at which key mechanosensing pathways are activated: integrins begin transmitting tension at around 5 pN, talin unfolding exposes cryptic vinculin-binding sites at 5–15 pN [17, 18]. While forces of this magnitude may not be sufficient to instantaneously extract a transmembrane protein from a fluid lipid bilayer, which typically requires > 100 pN [19], they are large enough to significantly accelerate the rate of extraction under sustained loading. This has direct implications for processes such as viral disassembly, where even a modest increase in the mechanical stress on surface proteins can compromise envelope integrity and stability. More generally, the fact that nanoparticle-exerted forces operate in the regime sensed by cellular mechanotransduction machinery suggests that specifically-tuned multivalent constructs could trigger downstream biological signalling, achieving downstream effects beyond those that can be produced with receptor engagement alone.

Before concluding, we would like to add some consideration about how this model can be adapted to treat other potential cases of interest, in particular, the important case where receptors are not randomly distributed but follow some geometrical pattern. In all the previ-

ous calculations, a surface with a random disposition of receptors is assumed by setting its average value on the surface $\sigma(r) = \sigma$, which also allows for the use of rotational and translational symmetry on the plane. However, the case of an ordered lattice of receptors can be equally studied by considering that more generally the receptor density is $\sigma = \sigma(\mathbf{r})$ and replacing the integrals in Eqs. 7, 8 with their appropriate values for non-uniform densities by setting $\sigma(\mathbf{r}) = \sum_{i=1}^{N_R} \delta(\mathbf{r} - \mathbf{r}_i)$, where $\mathbf{r}_i$ is the position of receptor i . Notice that in this case however, the exact particle position matters because of the loss of isotropicity in the plane where receptors are grafted; this and should be taken into account when calculating the bound partition (or the probability for the particle to bind at a specific spot) by not only sampling over all potential values of z , but also $\mathbf{r}_{NP}$ describing the lateral position of the nanoparticle. We do not treat this case here because it only complicates the mathematical description without adding any particular physical insight into the problem, and qualitatively the same trends are expected. Similarly, one could think about using accurate Monte Carlo simulations to numerically estimate the exact distribution of ΔG^{cnf} for a specific ligand-receptor pair or to include its dependence on the curvature of the grafting surface. The latter could also introduce additional effects due to its influence on the repulsive energy contribution. Whereas these effects will all change the results quantitatively, they will not change the underlying physics of the problem. In particular, the observed multivalent effects due to the competition between different bonding configurations, which translate to highly non-linear and non-monotonic behaviour of the force as a function of the various design parameters, will remain the same. For this reason, whereas simulations or theoretical extension/specializations might be necessary to treat a specific system quantitatively, we expect the general guidelines described here to maximise (or minimise) the exerted force to be robust to these details.

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Supplementary Information for the paper: Statistical mechanical theory of binding-induced stresses and consequences for virucidal activity of nanoparticles

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(Note: For consistency with the main text, it should be confirmed whether Daniele's affiliation also includes Imperial College London, since it is included in the main text.)

SI. DERIVATION OF THE FRACTION OF BOUND ADSORPTION SITES

SII. VERTICAL FORCE COMPONENT

Figure S1 and Figure S2 show the vertical (perpendicular to the surface) force component F_z^* , computed with the same parameters used for the radial force curves presented in Figure 2 and Figure 3 of the main text, respectively.

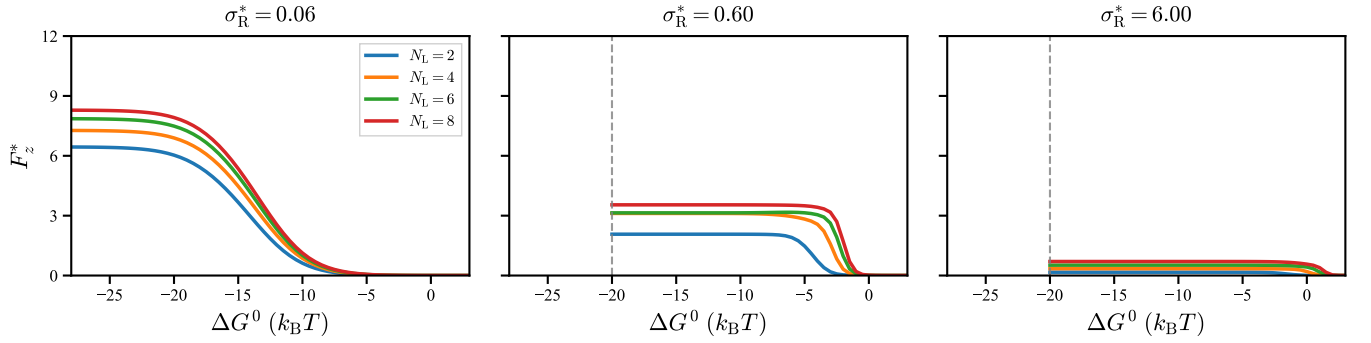


Figure S1: **Average vertical force on a bound receptor as a function of binding strength.** Curves computed using same parameters reported in caption of Figure 2 in the main text for the dimensionless average vertical force F_z^* . As noted in the main text, the qualitative behaviour is identical to that of the radial component F_r^* .

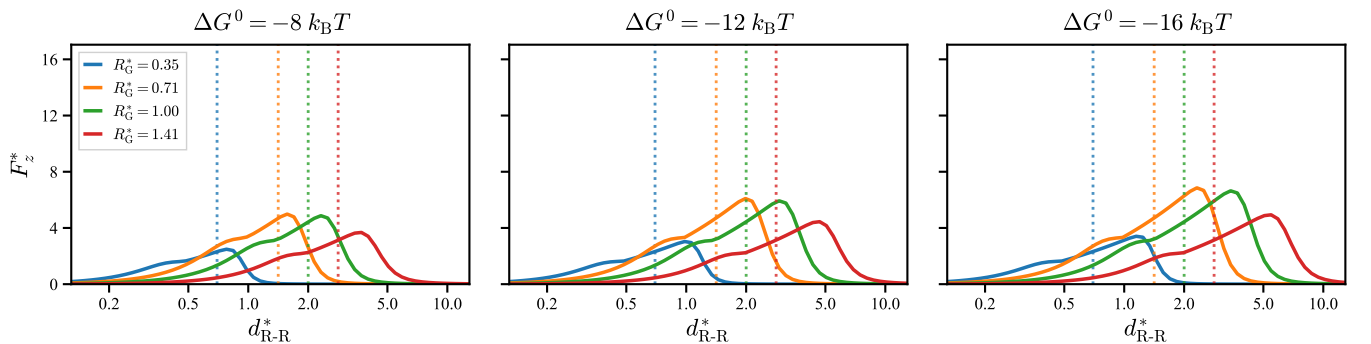


Figure S2: **Vertical force on a bound receptor as a function of receptor-receptor distance.** Curves computed using same parameters reported in caption of Figure 3 in the main text for the dimensionless average vertical force F_z^* . As for the radial component, the force is non-monotonic, peaking at an intermediate receptor-receptor spacing and vanishing in both the closely-packed and dilute limits.

SIII. EFFECT OF ALTERNATIVE NON-SPECIFIC REPULSION

In the main text, the repulsion interaction $A_{\text{steric}}(z)$ was modelled through the steric interaction of inert polymer chains grafted to the nanoparticle. To verify that the qualitative conclusions do not depend on the specific functional form chosen, we repeated the analysis using a generic short-range repulsion of the form

$$A_{\text{steric}}(z) = \frac{A_0}{z^2 + z_c^2}, \quad (\text{S1})$$

where A_0 controls the repulsion strength and z_c is a short-distance cutoff that prevents divergence at contact. This $1/z^2$ form captures the essential physics of short-range, monotonically decaying repulsion without reference to a specific polymer model.

Figure S3 shows the resulting dimensionless force components as a function of binding strength ΔG^0 for several values of the dimensionless repulsion amplitude $A_0^* = A_0/(k_B T R_G^2)$. The qualitative behaviour is identical to that found for the polymer shell model (Figure 4 of the main text): the radial force F_r^* decreases monotonically with increasing repulsion strength across the entire range of ΔG^0 , while the vertical force F_z^* exhibits a crossover—enhanced at strong binding and suppressed at weak binding. The total force magnitude $|\mathbf{F}^*|$ inherits this crossover. These results confirm that the design principles discussed in the main text are robust with respect to the detailed form of the non-specific repulsion.

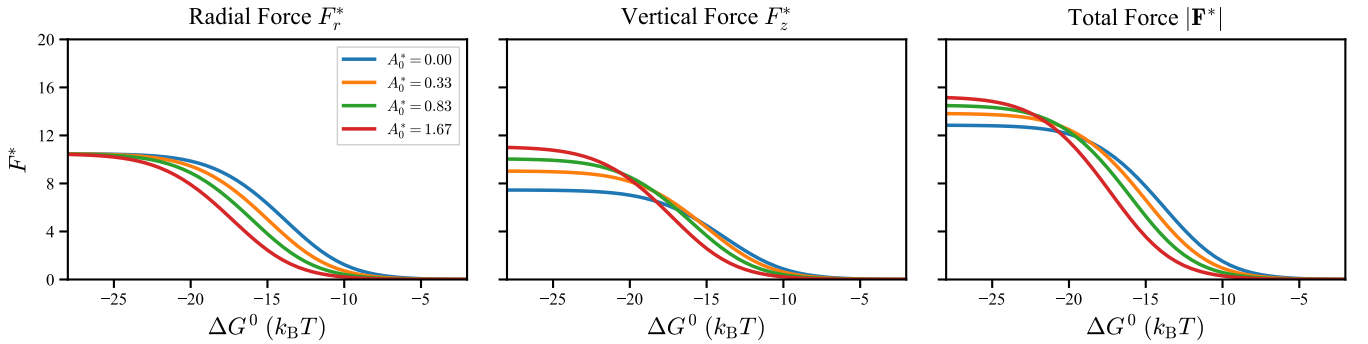


Figure S3: **Force on a bound receptor for a generic $1/z^2$ repulsive interaction.** Dimensionless radial F_r^* , vertical F_z^* , and total $|\mathbf{F}^*|$ force as a function of binding strength ΔG^0 for different values of the dimensionless repulsion amplitude A_0^* . Parameters: $\sigma_R^* = 0.06$, $z_c^* = z_c/R_G \approx 0.13$. The qualitative behaviour is the same as for the polymer shell model (Figure 4, main text).